Exp - 4

Study of Local Areas Network using a Switch & Ethernet Cables.

Aim:

Setup and configure a LAN using Switch and Ethernet cables in your lab.

Procedure:

Step 1: Plan and Design an appropriate network topology taking into account network requirements and equipments location.

Step 2: You can take 4 computers, a switch with 8, 16 or 24 ports which is sufficient for networks.

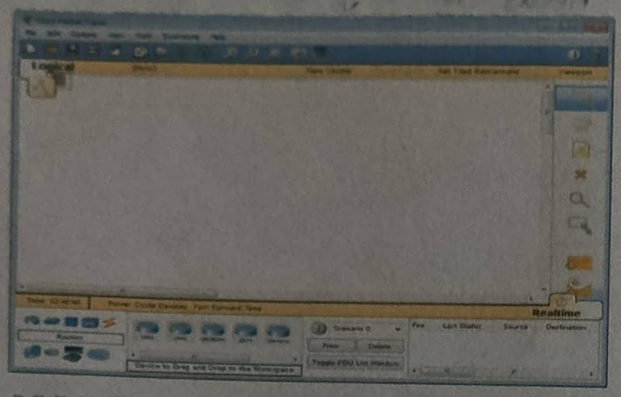
Step 3: Connect your computer to network switch via Ethernet cable, which is as simple as plugging one end of Ethernet cable into your computer.

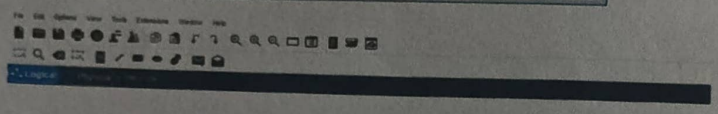
Step 4: Assign IP address to your PCs

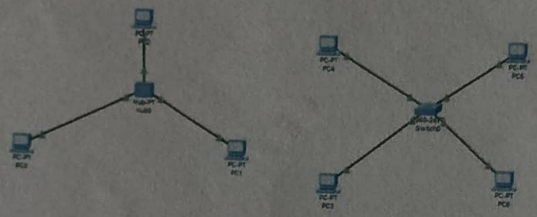
1. Log on to the client computer as Administrator or as User
2. Click Network and Internet connection.
3. Right click Local Area connection / Ethernet → Go to properties → select Internet Protocol → click on properties → select use the following ip address option & assign ipaddress.

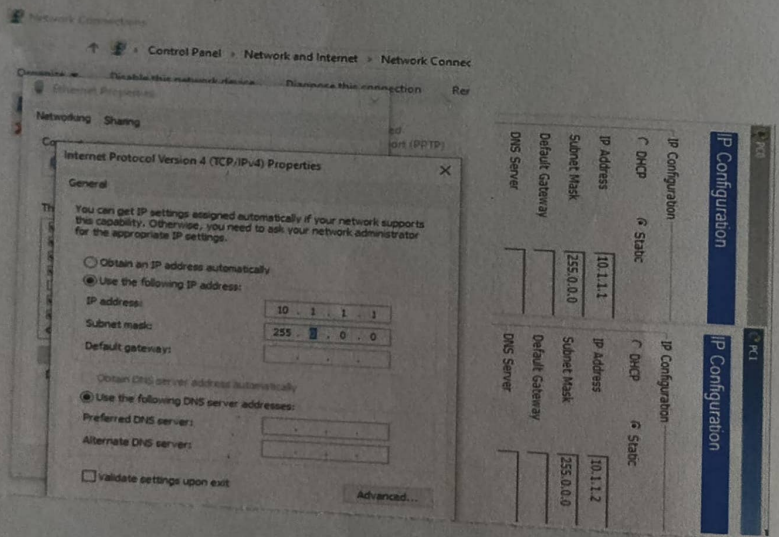
Step 5: Configure a network switch:

1. Connect your computer to the switch:
2. Log in to the web Interface
3. Configure basic settings
4. Assign IP address


Realtime






Control Panel
Network and Internet
Network Connec
Networking
Sharing
Internet Protocol Version 4 (TCP/IPv4) Properties
General
You can get IP settings assigned automatically if your network supports this capability. Otherwise, you need to ask your network administrator for the appropriate IP settings.
Obtain an IP address automatically
Use the following IP address:
IP address:
10 . 1 . 1 . 1
Subnet mask:
255 . 0 . 0 . 0
Default gateway:
Obtain DNS server address automatically
Use the following DNS server addresses:
Preferred DNS server:
Alternate DNS server:
Validate settings upon exit
Advanced...
IP Configuration
DHCP
Static
IP Address
10.1.1.1
Subnet Mask
255.0.0.0
Default Gateway
DNS Server
IP Configuration
DHCP
Static
IP Address
10.1.1.2
Subnet Mask
255.0.0.0
Default Gateway
DNS Server

Step 6: check the connectivity between switch & other machine by using ping command in the command prompt of the device.

Step 7: Select a folder → go to properties → click sharing tab → share it with everyone on the same LAN

Step 8: Try to access the shared folder others computer of the network.

Result:

Thus, a local area network (LAN) was successfully set up and configured using a network switch & ethernet cables.